

Is Seattle rainier than Portland, Maine?

Seattle has been infamous as one of the rainiest cities in the US, especially on the West Coast. Therefore, I wanted to choose Portland, Maine, farther along the East Coast, to see whether Seattle still holds that reputation.

In General, the data indicates that the two cities have very different rain patterns. Particularly, Seattle gets more frequent light rain, which is concentrated in the winter. However, Portland gets fewer, heavier storms spread through the year.

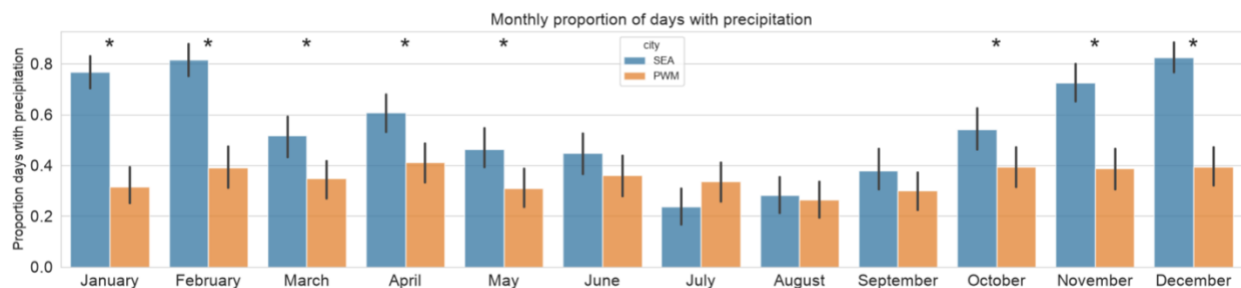
Data and method

Daily precipitation data were extracted from NOAA's Climate Data Online, which is in the public domain. Seattle has one station, which is SEATTLE 2.1 ESE (US1WAKG0225). Portland has more than 29 stations; therefore, we subjectively selected Portland International Jetport station (USW00014764) for our comparison purposes. The dataset covers the period from January 2018 to December 2022.

The dataset covers the period from January 2018 to December 2022. The dataset was processed and cleaned up before the analysis was performed, including format editing and unusable data removal. Both datasets have all 1,826 days. Then, the two cities were joined into one tidy table of 3,652 rows.

The data for Seattle has 190 days with missing data, including 168 are not available from the file, and 22 days have a blank value. Each missing day was filled automatically with Seattle's average for the same calendar date in the other years.

How often it rains



This chart indicates the clearest rain pattern difference between Seattle and Portland, Maine. In general, Seattle has significantly more rainy days in 8 of the 12 months from October through May. Particularly, Seattle has rain on 80% of days compared to Portland, which is 40% during that period. From June to September, Seattle's dry season brings it down to Portland's level.

When the rain falls

SEASON	SEATTLE	PORTLAND
Winter (Dec – Feb)	46.2%	25.7%
Spring (Mar – May)	19.1%	21.1%
Summer (Jun – Aug)	7.1%	22.5%
Fall (Sep – Nov)	27.6%	30.7%

Almost 50% of Seattle's precipitation is in Winter, whereas Portland's precipitation spreads evenly during the year.

What a rainy day looks like

MEASURE	SEATTLE	PORTLAND
Median Rainy Day	0.10 in	0.18 in
Wettest Single Day	2.60 in	3.40 in
Longest Wet Streak	36 days	6 days
Longest Dry Streak	34 days	17 days

Portland's rainy days are heavier (more than 1 inch) on average. Meanwhile, Seattle's rainy and dry seasons last longer

Conclusions

- Portland is rainier than Seattle by the amount. It had about 6 more inches per year and had three times as many days with more than an inch.
- Seattle is rainier than Portland by frequency. It had rain on about half of all days. In addition, Seattle has a higher number of rainy days in 8 of the 12 months compared to Portland.
- In conclusion, Seattle has wetter winters and drier summers, while Portland's precipitation is steady all year.